

CONTACT
INFORMATION

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EDUCATION

National University of Singapore, Singapore.

2025–now

◦ Doctor of Philosophy, Computer Science.

Supervisor: Prof. HE Bingsheng

National University of Singapore, Singapore.

2023–2025

◦ Master of Computing, Computer Science Specialization.

GPA: 4.4/5.0

South China University of Technology (SCUT), Guangzhou, China.

2019–2023

◦ B.Eng., Software Engineering.

GPA: 3.6/4.0

PRIZES
AND
AWARDS

- **Excellent Degree Dissertation of South China University of Technology** 2023
- **Honorable Mention** in Mathematical Contest in Modeling 2023
- **National Scholarship** 2022
- **Bronze Medal (46th)** in ICPC Asia-East Continent Final(Xi An) 2022
- **101/1608** in CCF-DBCI Competition of "Small Sample Data Classification" 2022
- **Silver Medal (46th)** in ICPC Asia Regional Contest(Ji Nan) 2021
- **44/3567** in CCF-DBCI Competition of "Recognition of figure skaters' skeleton points based on Paddle" 2021
- **First Prize** in National Olympiad in Informatics in Province(NOIP) 2017

PREPRINTS
AND
PUBLICATIONS

- **CHEN Han**, Zining Zhang, Wenqi Pei, Bingsheng He, Ming Wu, Jason Zeng, Michael Heinrich, Wei Wu, Hongbao Zhang. **MemForest: An Efficient Agent Memory System with Hierarchical Temporal Indexing**. Under review at VLDB 2027 2026
- Wenqi Pei, Shizheng Hou, Boyan Li, **CHEN Han**, Zhichao Shi, Yuyu Luo. **ROSE: An Intent-Centered Evaluation Metric for NL2SQL**. Accepted at ACL 2026 Main 2026
- **CHEN Han**, Tao Huang, Phuong Pham, Shuang Wu. **HiAE: A High-Throughput Authenticated Encryption Algorithm for Cross-Platform Efficiency** 2025
- **CHEN Han**, Zicong Jiang, Zining Zhang, Bingsheng He. **LogQuant: Log-Distributed 2-Bit Quantization of KV Cache with Superior Accuracy Preservation**. Accepted at ICLR 2025 Workshop on Sparsity in LLMs 2025
- Wenqi Pei, Hailing Xu, Henry Hengyuan Zhao, **CHEN Han**, Zining Zhang, Shizheng Hou, Luo Pingyi, Bingsheng He. **Optimizing Small Language Models for NL2SQL**. Accepted at ICLR 2025 Third Workshop on Deep Learning for Code 2025

PROJECT
EXPERIENCE

- **Ph.D. Research: Efficient Agent Memory System for LLM Agents** in **National University of Singapore**
Supervisor: Prof. HE Bingsheng Jan.–May 2026
 - Design *MemForest*, an agent memory system that reformulates LLM agent memory as a write-efficient temporal data problem, decoupling memory construction into parallelizable operations.

- Propose *MemTree*, a hierarchical temporal index that organizes memory as time-ordered trees with localized per-node updates, replacing costly full-state rewrites.
 - Achieve 89% pass@8 accuracy on LongMemEval-S with approximately $6\times$ higher memory construction throughput than state-of-the-art EverMemOS.
 - Sponsored by 0G Labs (Zero Gravity Labs), a blockchain infrastructure company building the decentralized L1 for AI agents.
 - Under review at VLDB 2027. [\[Paper\]](#) [\[Code\]](#)
- ***Cryptography Engineer: Efficient Cipher Implementation and Whitebox Design* in SG Digital Trust Lab, Singapore Research Center, 2012 Laboratory**

Mentor: [Dr. WU Shuang](#) Jan.-June 2025

 - Design whitebox implementation of AES for different platforms, products and scenarios.
 - Implement HiAE cipher for kernel and user space, integrate it into several products, including mobile phone, server and auto-driving chips.
 - Optimize the performance of SM2, SM3 and SM4 ciphers on ARM platforms.
- ***Research Assistant: optimization for large language Model inference* in National University of Singapore**

Mentor: [Prof. HE Bingsheng](#) May-Sept. 2024

 - Design a new 2-bit KV Cache quantization for LLMs base on attention patterns. achieve over 200% accuracy improvement at same compression rate.
 - Implement a adaptive API for popular inference frameworks like Python' s `transformers`, boosts batch size by 60% without increasing memory consumption.
 - Accepted at ICLR 2025 Workshop on Sparsity in LLMs. [\[Paper\]](#) [\[Code\]](#)
- ***Internship: Cryptography Engineer* in SG Digital Trust Lab, Singapore Research Center, 2012 Laboratory**

Mentor: [Dr. HUANG Tao](#) Jan.-Dec. 2024

 - Design a 'XAXX' structure to efficiently utilize the distinct pipelines of both ARM and x86 (with AES-NI) architectures, achieving high IPC.
 - Build a new AEAD (Authenticated Encryption with Associated Data) cipher named '*HiAE*' based on the 'XAXX' structure, which is $5\times$ faster than AES-256-GCM across different platforms and outperforms all existing AEAD ciphers on latest ARM and x86 processors.
 - Create a new record of 340Gbps throughput for single-threaded and single-stream AEAD encryption. Applied to the various products.
 - Open-source on Cryptology ePrint Archive [\[Paper\]](#) [\[Code\]](#)
- ***Research Assistant: Parallel Graph Partitioning* in Hong Kong University of Science and Technology, Guangzhou**

Mentor: [Prof. Zeyi WEN](#) Apr-Sept. 2023

 - Implement a library of parallel hyper graph partitioning with openMP task.
 - Realize a multiple node contraction algorithm for graph coursenning.
 - Research parallel graph partitioning algorithm and adapt it for fill-in reduction of sparse matrix.
- ***Symmetric Matrix Solving Algorithm Parallel Optimization for ARM Architecture***

Mentor: [Prof. TANG Deyou](#) May-Dec. 2022

 - Optimize and parallel Bounded Bunch-Kaufman Algorithm(*sysv_rk subroutine of LAPACK) for solving symmetric matrix on ARM server processor with NEON instruction set and openMP.

- Implement a parallel column reordering method in row swap of solving symmetric matrix to enhance memory access locality for column major matrix for better cache hit rate and parallelism, achieving a performance improvement from 320Gflops to 580Gflops.
- Implement the same optimization on Skylake Intel processor and achieve 2-5x multi-core speedup than MKL library for *sytrs_3 subroutine of LAPACK.
- Awarded as the Excellent Degree Dissertation of South China University of Technology.

TECHNICAL SKILLS

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- *English*: IELTS(6.5), CET-4, CET-6.
 - *Programming Languages*: C/C++, Fortran, p4-16, Python, SQL, L^AT_EX.
 - *Technical Skills*: openMP, SIMDs(NEON, AVX512), MPI, PyTorch, CUDA.
 - *TestDemo Certificate*: C++, TOP 10%, LINUX, TOP 10%, PYTHON, TOP 10%.
 - *Kaggle Certificate*: Data Visualization, Intro to Machine Learning, Intro to Deep Learning, Intro to Game AI and Reinforcement Learning.
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EXCHANGE EXPERIENCE

- **Online Academic Program on Machine Learning, McGill University** Jan.-Feb. 2022

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Address: NUS School of Computing, 13 Computing Drive, Singapore 117417



教育经历

新加坡国立大学, 新加坡	2025–现在
◦ 计算机科学博士.	导师: 何丙胜教授
新加坡国立大学, 新加坡	2023–2025
◦ 计算机科学硕士.	GPA: 4.4/5.0
华南理工大学, 广东省广州市	2019–2023
◦ 工学学士, 软件工程专业.	GPA: 3.6/4.0

获奖荣誉

• 华南理工大学本科优秀毕业设计 (论文)	2023
• 二等奖美国大学生数学建模竞赛 (MCM/ICM)	2023
• 铜牌第 46 届 ICPC 国际大学生程序设计竞赛亚洲区决赛	2022
• 101/1608 CCF-DBCI ”小样本数据分类算法” 竞赛	2022
• 国家奖学金	2022
• 银牌第 46 届 ICPC 国际大学生程序设计竞赛 (济南站)	2021
• 44/3567 CCF-DBCI ”基于飞浆实现花样滑冰选手骨骼点识别” 竞赛	2021
• 一等奖全国青少年信息学奥林匹克联赛 (NOIP)	2017

预印本和发表
论文

• <i>CHEN Han, Zining Zhang, Wenqi Pei, Bingsheng He, Ming Wu, Jason Zeng, Michael Heinrich, Wei Wu, Hongbao Zhang. MemForest: An Efficient Agent Memory System with Hierarchical Temporal Indexing.</i> 投稿至 VLDB 2027 (审稿中)	2026
• <i>Wenqi Pei, Shizheng Hou, Boyan Li, CHEN Han, Zhichao Shi, Yuyu Luo. ROSE: An Intent-Centered Evaluation Metric for NL2SQL.</i> 录用于 ACL 2026 Main	2026
• <i>CHEN Han, Tao Huang, Phuong Pham, Shuang Wu. HiAE: A High-Throughput Authenticated Encryption Algorithm for Cross-Platform Efficiency</i>	2025
• <i>CHEN Han, Zicong Jiang, Zining Zhang, Bingsheng He. LogQuant: Log-Distributed 2-Bit Quantization of KV Cache with Superior Accuracy Preservation.</i> Accepted at ICLR 2025 Workshop on Sparsity in LLMs	2025
• <i>Wenqi Pei, Hailing Xu, Henry Hengyuan Zhao, CHEN Han, Zining Zhang, Shizheng Hou, Luo Pingyi, Bingsheng He. Optimizing Small Language Models for NL2SQL.</i> Accepted at ICLR 2025 Third Workshop on Deep Learning for Code	2025

项目经历

• 博士研究: 面向大语言模型智能体的高效记忆系统: 新加坡国立大学	2026/01–2026/05
导师: 何丙胜教授	
◦ 设计了 <i>MemForest</i> 智能体记忆系统, 将 LLM 智能体记忆重新建模为写入高效的时序数据问题, 将记忆构建解耦为可并行操作。	
◦ 提出 <i>MemTree</i> 层次化时序索引结构, 将记忆组织为时间有序的树, 用局部节点更新替代代价高昂的全状态重写。	

- 在 LongMemEval-S 基准上达到 89% pass@8 准确率, 同时记忆构建吞吐量约为最新方法 EverMemOS 的 6 倍。
 - 该项目由0G Labs (Zero Gravity Labs)赞助, 一家为 AI 智能体构建去中心化 L1 区块链基础设施的公司。
 - 投稿至 VLDB 2027 (审稿中). [\[Paper\]](#) [\[Code\]](#)
- 密码算法工程师: 高效密码算法实现和白盒设计:** 新加坡数字信任实验室, 华为 2012 实验室新加坡研究所 2025/01–2025/06
 导师: 吴双博士
 - 设计了不同场景和平台的 AES 白盒实现。
 - 在用户态和内核态实现了 HiAE 密码算法, 并集成到多种产品中, 包括手机、服务器和自动驾驶芯片。
 - 优化了 SM2、SM3 和 SM4 密码算法在 ARM 平台上的性能。
- 科研助理: 大语言模型推理优化:** 新加坡国立大学 2024/05–2024/09
 导师: 何丙胜教授
 - 设计了一种基于注意力模式的 2 位 KV Cache 量化方法, 在相同压缩率下, 提高了 200% 的准确率。
 - 实现了一个适应性 API, 用于流行的推理框架, 如 Python 的 transformers, 在不增加内存消耗的情况下, 将批处理大小提高了 60%。
 - 该项目已被 ICLR 2025 Sparsity in LLMs Workshop 接受。 [\[Paper\]](#) [\[Code\]](#)
- 实习生: 密码算法工程师:** 华为 2012 实验室新加坡研究所谢尔德实验室 2024/01–2024/12
 导师: 黄涛博士
 - 设计了一种新的 'XAXX' 算法结构, 可以高效利用 ARM 和 x86(带 AES-NI) 架构的流水线, 实现了高 IPC。
 - 基于 'XAXX' 结构构建了一种新的 AEAD(带关联数据的认证加密) 密码算法, 'HiAE', 在多种平台上相较 AES-256-GCM 提升 5 倍以上性能, 在最新的 ARM 和 x86 处理器上优于所有现有的 AEAD 密码算法。
 - 创造了单线程单流 AEAD 加密的 340Gbps 新纪录, 并应用于多种华为产品。
 - 该项目已在 Cryptology ePrint Archive 上开源。 [\[Paper\]](#) [\[Code\]](#)
- 科研助理: 香港科技大学广州校区** 2023/04–2023/09
 导师: 文泽忆教授
 - 实现了一个基于 openMP Task 的并行多层拓扑图分割库。
 - 在图压缩中实现了一种多节点收缩算法。
 - 研究将并行图分割算法, 应用于稀疏矩阵的填充减少。
- 对称矩阵函数求解 BBK 算法的并行优化** 2022/04–2022/12
 导师: 汤德佑教授
 - 在 ARM 处理器上利用 NEON 指令集和 openMP 对 Bounded Bunch-Kaufman 算法 (LAPACK 库 *sysv_rk 函数) 进行并行优化。
 - 实现了一种并行列重排方法, 在列优先矩阵的行交换中改进访存局部性, 使得缓存命中率和并行性能得到提高, 在鲲鹏 920-6426 处理器上的单精度性能从 320Gflops 提升到 580Gflops。
 - 将该方法移植到 Intel Skylake 处理器上, 对比 MKL 库的 *sytrs_3 函数, 实现了 2-5 倍的并行性能提升。
 - 该项目获评华南理工大学本科优秀毕业设计。

专业技能

- 英语认证水平: CET-4, CET-6, IELTS(6.5).
- 编程语言: C/C++, Fortran, p4-16, Python, SQL, L^AT_EX.

- 编程技能: openMP, SIMDs(NEON, AVX512), MPI, PyTorch, CUDA.
- *TestDemo* 编程技能认证: C++, TOP 10%, LINUX, TOP 10%, PYTHON, TOP 10%
- *Kaggle* 课程认证: 数据可视化, 机器学习, 深度学习, 强化学习

交换经历

-
- 机器学习线上访学项目, 麦吉尔大学 2022/01–2022/02